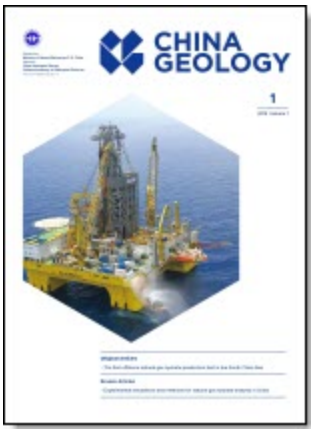
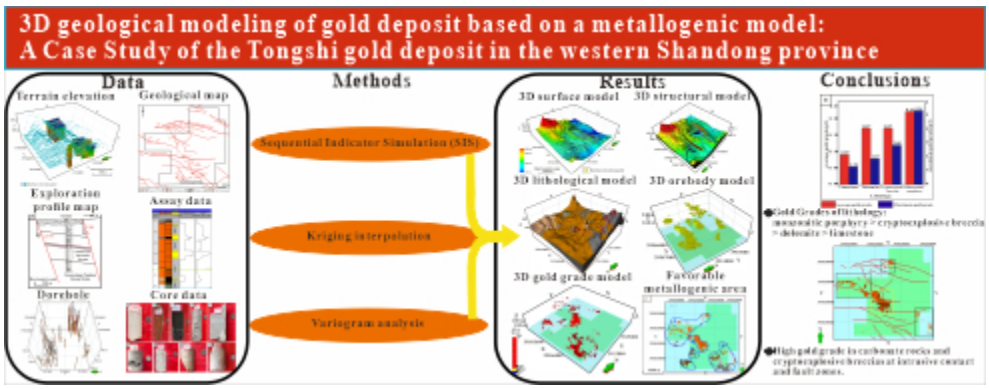


Three-Dimensional Geological Modelling of a Gold Deposit Using a Metallogenic Model: A Case Study of the Tongshi Gold Deposit in Western Shandong Province, China

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Graphical abstract

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Highlights

- First 3D modeling integrating PETREL with multi-source data for Tongshi gold deposit.
- Three mineralization models established: cryptoexplosive breccia, microfracture-disseminated, porphyry vein.
- Four prospective zones identified with stratoid orebodies in monzonitic porphyry-carbonate contacts.
- Gold mineralization is genetically linked to monzonite porphyry intrusion.
- NW/EW-trending fault systems control 83% of orebodies distribution in 3D space.

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Three-Dimensional Geological Modelling of a Gold Deposit Using a Metallogenic Model: A Case Study of the Tongshi Gold Deposit in Western Shandong Province, China

You-cai Shi [a], Cun-fei Ma [a, *], Wei Guo [b], Jin-qiang Tian [a], Peng Liu [b], Xin Zhang [b], Zhen-zhen Wang [c], Fu-yun Cong [a], Xian-xu Fang [a], Hong-bo Wang [a]

[a] School of Geosciences, China University of Petroleum (East China), Qingdao, Shandong 266000, China.

[b] Shandong Energy Group South America Company Limited, Qingdao, Shandong 266000, China.

[c] School of Earth Sciences and Resources, China University of Geosciences Beijing, Beijing 100000, China.

Abstract: The Tongshi Gold Deposit in western Shandong Province, China, is an area in which substantial progress has been made in identifying ore-controlling factors, gold mineral types, and metallogenic mechanisms. However, the three-dimensional(3D) spatial distributions of the gold deposits remain unclear, which significantly limits exploration and development. Guided by the metallogenic model, this study employed 3D modeling technology and integrated multidisciplinary data to construct a 3D geological model of the Liujiazhuang Beiling Tongshi Gold Deposit and to delineate prospective exploration areas. The results include: (1) metallogenesis was governed by magmatic activity, fault-fracture systems, and host stratigraphic units; (2) three mineralization models were established: crypto-explosive breccia-type, microfracture-disseminated, and porphyry-vein-type orebodies; (3) a 3D geological model was constructed through the integration of multiple data types, incorporating explicit and implicit modeling methodologies; (4) four potential prospecting areas that contain monzonite porphyry-altered and layered microfracture-disseminated orebodies were delineated, together with minor breccia-type orebodies.

Keywords: Tongshi Gold Deposit, Metallogenic model, 3D geological model, Mineralization favorable area.

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1. Introduction

With the continuous improvement in geological understanding of ore deposition and exploration technology, western Shandong has gradually emerged as a major old-producing province in Shandong after the Jiaodong region (Chen J et al., 2024; Goldfarb RJ and Santosh M, 2014; Xie SF et al., 2022). Until now, studies of gold deposits in the Liujiazhuang Beiling area of Pingyi County have relied on two dimensional (2D) approaches, including magnetic anomaly interpretation and resistivity interpretation of geological units. These studies have not addressed the occurrence of orebodies in three dimensions (3D), which is needed to enhance further exploration (De Oliveira SB and Bertossi LG, 2023; Huang JN et al., 2010). Consequently, it is necessary to construct a 3D geological model to adequately elucidate the orebody properties, delineate the distribution of gold grades, and predict orebody occurrences.

The Tongshi Gold Deposit in the Liujiazhuang Beiling area of Pingyi County, Shandong Province, is located in the southwestern part of the Tongshi Complex, which is a subvolcanic intrusion composed of diorite porphyrite and monzonite porphyry (Fig.1). Several previous studies in the area were conducted in two dimensions (2D) (Aykaç S et al., 2022; Huang JN et al., 2010; Kalscheuer T et al., 2018). However, with the development of 3D modeling technology, 3D geological modeling to predict the occurrences of concealed orebodies has become a primary focus of mineral exploration in recent years (Chen JP et al., 2014; Geng YY et al., 2024; Liu LM et al., 2016; Mao XC et al., 2024; Xiang J et al., 2020; Zhao ZH et al., 2024). For example, Payne CE et al. (2015) integrated geological and geophysical data to construct a 3D geological model of hydrothermal gold deposits in New Zealand's Taupo Volcanic Zone, indicating that showed high exploration potential in a significant portion of the deep area. Arias M et al. (2021) utilized ArcGIS 10.8 (Esri) and GeoModeller 4.9 (BRGM/Intrepid Geophysics) for 3D geological modeling of the Touro Cu deposit in the NW Iberian Peninsula, delineating orebody distribution and mineralized zones. Enemark T et al. (2024) developed a probabilistic 3D modeling method combining geological and geophysical data that offers innovative exploration approaches. Zhu X et al. (2021) used 3D modeling technology to predict the favorable metallogenic areas of the Gejiu tin deposit in Yunnan, China, demonstrating the effectiveness of 3D modeling technology in predicting the occurrences of orebodies. Huang YZ et al. (2023) applied 3D visualization to the Zimu-dang gold deposit in southwestern Guizhou Province, China, which provided significant guidance for prospect prediction. Collectively, these case studies demonstrate that the 3D modeling technology is effective for orebody prediction.

Currently, ore deposit structures (Du XB et al., 2022), ore-forming fluids (Hu HB et al., 2006), the metallogenic models (Yu XF et al., 2022; Zhang YM et al., 2024), gold-bearing mineral types (Sun et al., 2019), and metallogenic predictions (Chen YQ and Liu HG, 2000; Zhang YQ et al., 2010) were elucidated for the Tongshi Gold Deposit. However, the 3D spatial distributions of strata, structures, and orebodies remain unclear. In this study, a 3D geological model of the

Tongshi Gold Deposit was constructed using the integration of multiple types of geological and geophysical data, incorporating explicit and implicit modeling methodologies. Traditional ore deposit geological modeling typically employs either explicit or implicit modeling in isolation (Liu LM et al., 2016; Liu ZK et al., 2021; Vollgger SA et al., 2015). The 3D geological model constructed in this study using a hybrid explicit-implicit approach demonstrated superior accuracy compared with conventional modeling methods. Based on this model, spatial distribution analysis and orebody prediction were performed to identify the prospective areas.

2. Geological setting

The study area is located west of the northern section of the Yangan fault and southwest of the Tongshi complex (Yin TT et al., 2019; Yu XF, 2010; Yu XF et al., 2009; Zhao BB and Chen YQ, 2011; Fig. 1). The Neoarchean intrusive rocks and the Neoarchean Taishan Group Shancaoyu Formation constitute the crystalline basement overlain by Cambrian and Quaternary sedimentary cover. The strata in the mining area exhibited a predominant NW-SE strike with NE dip directions of 3°–20°. These include the Neoarchean Taishan Group, Cambrian Changqing Group (Zhushadong and Mantou Formations), Ordovician Jiulong Group (Zhangxia Formation), and the Quaternary Piedmont and Yihe Group (Yin TT et al., 2019). According to core observations, the lithologies are primarily monzonite granite, granodiorite, monzonite porphyry, monzonite diorite porphyrite, plagioclase amphibolite, breccia, calcareous dolomite, argillaceous limestone, and gravelly sand and clay (Fig. 2).

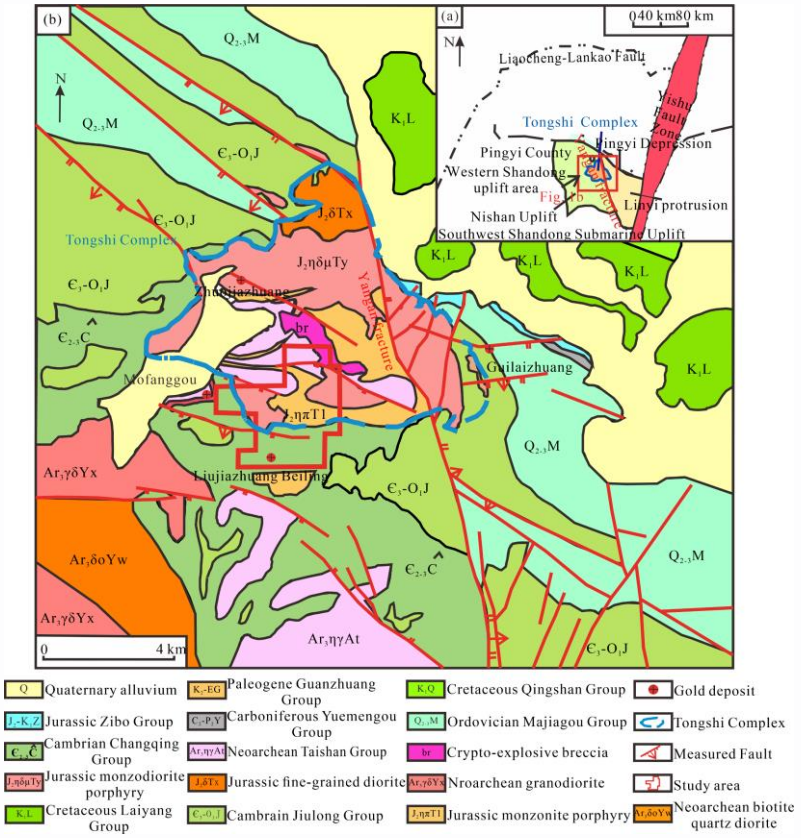


Fig. 1. Regional geological map of Tongshi Gold Field (altered from Zhang et al., 2024).



Fig. 2. Photographs of cores in the study area. (a) Fine-spotted monzonite porphyry; (b) Medium-spotted monzonite porphyry; (c) Plagioclase amphibolite; (d) Monzonite granite; (e) Calcareous dolomite; (f) Argillaceous limestone; (g) Fluoritized medium- to fine-spotted monzonite porphyry; (h) Coarse-spotted monzonite porphyry; (I) Breccia.

The structure in the mining area is fault-controlled, with a total of 25 faults (Fig. 3). These faults are categorized into three groups: NW-trending, sub-E–W-trending, and NE-trending systems. Among these, the NW-trending faults exhibit the highest spatial density and structural continuity, and probably served as primary conduits for hydrothermal fluids migration and ore deposition. Most NW-trending faults are pre-mineralization features that also provided migration channels and structural traps for ore-bearing hydrothermal fluids. Sub-E–W-trending faults, such as the Yingziwa Fault, exhibit tensile characteristics. The NE-trending faults are high-angle extensional faults that significantly influence the surface distribution of the Cambrian Changqing Group strata. Fault surfaces display planar geometries characterized by prominent slickenside striations and mirror surfaces with slicken-fibers, interpreted as post-mineralization reactivation features (Zhang YM et al., 2024).

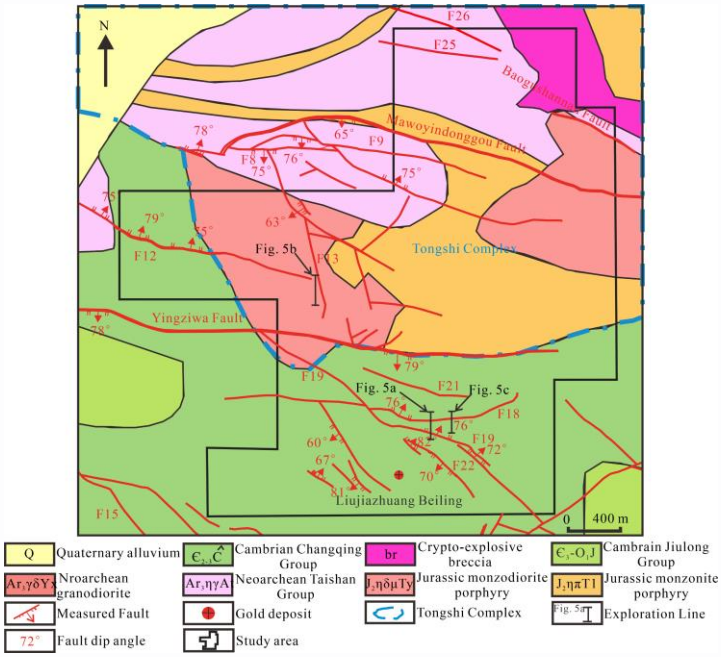


Fig. 3. The fault plane distribution map of the study area.

The magmatic rocks in the study area span Neoarchean to Mesozoic. Neoarchean monzonite granite is distributed in the northeast of the study area, while the Mesozoic monzonite porphyry and monzodiorite porphyry are widely distributed (Fig. 1). Wall-rock alteration includes fluorite alteration, kaolinization, pyritization, carbonatization, silicification, and chloritization. Among these, the occurrence of silicification and fluorite alteration exhibit the strongest correlation with gold mineralization (Shen K et al., 2001; Yin SZ et al., 2001).

3. Metallogenic conditions and model of the ore deposit

Multi-stage magmatic intrusions and frequent tectonic activity in the study area have created favorable metallogenic conditions for gold deposit formation. During the Middle Jurassic, multi-stage tectonic events generated NE-trending-, NW-trending-, and sub-E–W-trending secondary structural systems that established key structural conduits for ore-bearing hydrothermal systems (Zhang YM et al., 2024), while dolomitic limestone, argillaceous limestone, and banded dolomite in the Middle to Upper Cambrian to Lower Ordovician provided optimal lithological traps for gold accumulation (Zhang YQ et al. 2010).

The Tongshi Gold Deposit is classified as a low-temperature epithermal gold deposit (Niu CY et al., 2018; Wang JX et al., 2015; Yu XF et al., 2022). A preliminary mineralization model was established by combining the geological characteristics and ore-controlling factors of the deposit with previous research and analysis (Zhang YM et al., 2018; Fig. 4). In weak lithospheric zones, high fluid pressures favored subvolcanic crypto-explosive events. The host rocks within these tectonic zones were fragmented and transported along faults with ore-bearing hydrothermal fluids, forming near-surface crypto-explosive breccia-type orebodies. Hydrothermal fluids percolating through the microfracture networks reacted with limestones in the Zhangxia Formation to form microfracture-disseminated orebodies. In contrast, when mineral-laden fluids encounter chemically inert consolidated rocks, metal precipitation occurs within the vein systems, forming porphyry-vein-type orebodies (Zhang YM et al., 2021).

Analysis of geological cross-sections and composite well logs indicates that the orebodies predominantly occur as stratiform-type geobodies, either overlying or intercalated within the monzonite porphyry intervals (Fig. 5). Consequently, this study is focused on developing a 3D geological model for porphyry-hosted alteration gold orebodies occurring in layered monzonite porphyry and hydrothermally controlled layered microfractured gold orebodies.

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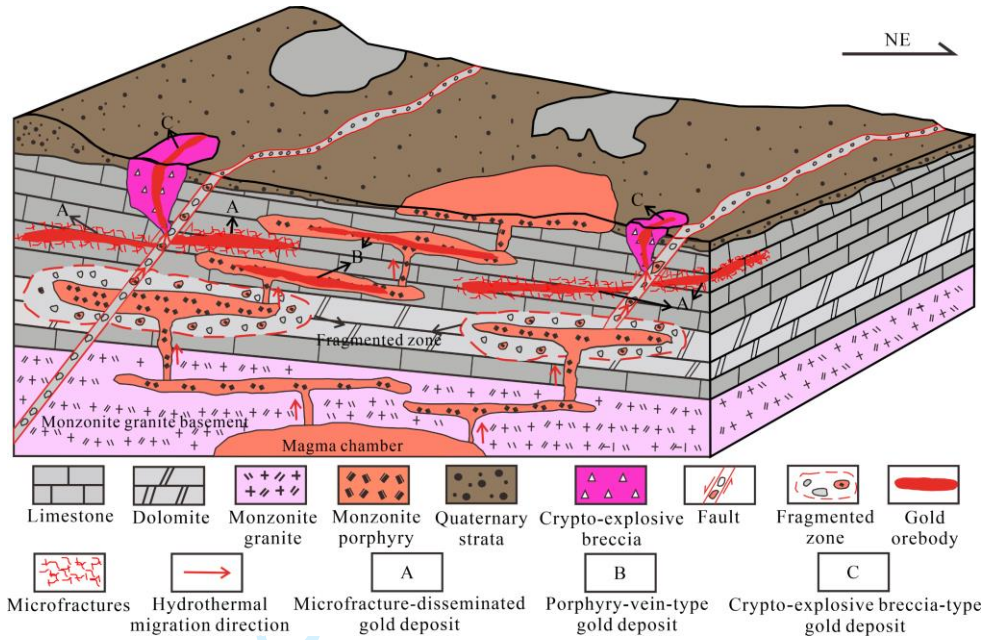


Fig. 4. Metallogenic model for gold deposit in the study area.

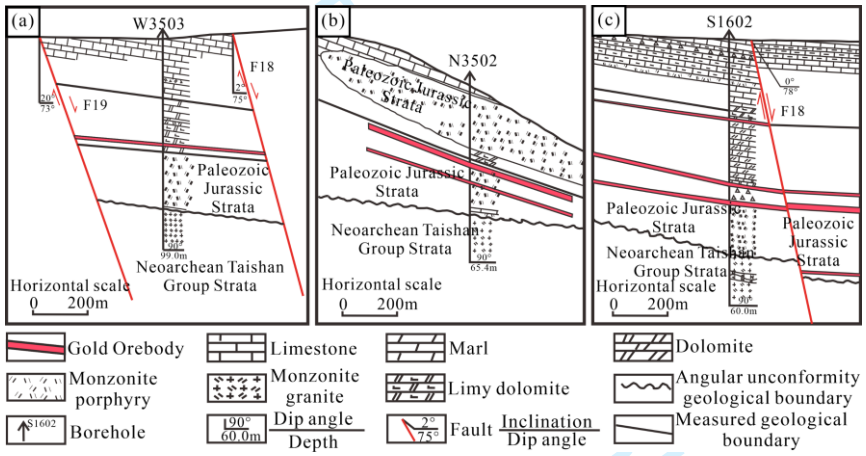


Fig. 5. Schematic diagram of 2D cross-sections in the study area.

4. Data and methods

4.1. Database development

Complete and accurate data are critical parameters for accurate modeling (Guo FS et al., 2020; Kaufmann O and Martin T, 2008; Wang GW et al., 2011). The dataset in this study contained: (1) a 1:5000-scale geological and topographic map with a magnetic survey overlay (4 sheets); (2) 1:1000-scale 2D exploration cross-sections (17 sections); (3) 1:2000-scale orebody resource estimation horizontal projection maps (four sheets); (4) 89 borehole columnar sections; (5) controlled source audio-frequency magnetotellurics (CSAMT) inversion resistivity cross-sections (10 profiles); (6) transient electromagnetic method (TEM) inversion resistivity cross-sections (12 profiles); (7) 4177 surface elevation points; (8) 6338 gold grade assay samples; and (9) a metallogenic model of the study area. These data comprise the geological data and metallogenic model required to build a 3D surface model and construct a tectonic framework

model (Chen J et al., 2023; Deng J et al., 2020; Huang JX et al., 2023; Jin X et al., 2020; Mao XC et al., 2024), and establish the orebody model and the gold grade model.

A detailed geological database is necessary to perform 3D geological modeling and visualization-driven quantitative prediction of deep orebodies (Chen KQ et al., 2024; Liu JK et al., 2025; Mao XC et al., 2024). The geological and topographic maps, exploration cross-sections, and resource-level projection maps were digitized and digitally processed to build a geological database. The borehole logs were digitally compiled to construct an exploration engineering database (Fig. 6). The surface topography and geophysical interpretation results were imported into the Petrel™ E&P software platform (developed by Schlumberger, Houston, Texas, USA) to construct the geophysical inversion database (Fig. 7).

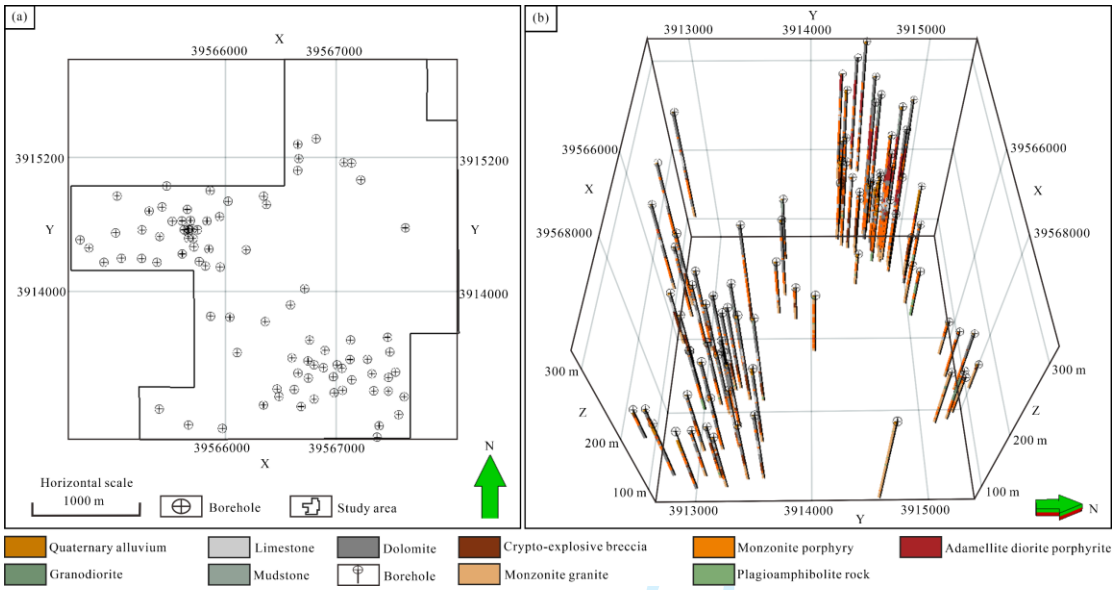


Fig. 6. Lithology logs in the exploration engineering database derived using Petrel™ E&P software. (a) Borehole planar distribution map; (b) 3D visualization of borehole data.

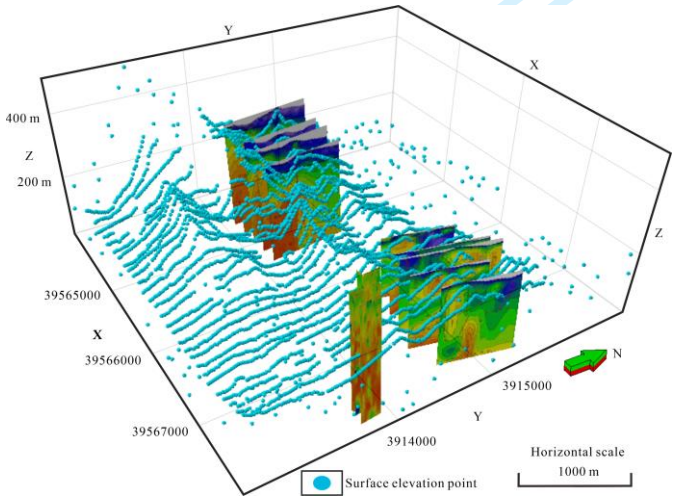


Fig. 7. Map of geophysical inversion database results derived using Petrel™ E&P software. Surface elevation points are integrated to model surface morphology. Seismic slices are utilized to interpret various geological bodies.

4.2. Methodology

3D geological modeling integrates geology and geostatistics to synthesize surface geological data, borehole data, and geophysical datasets (Geng YY et al., 2024; Li B et al., 2022; Wang GW et al., 2012) to reconstruct subsurface geological bodies and orebodies. The established geological and metallogenic models indicate that gold mineralization in the study area exhibits a strong spatial correlation with the stratified monzonite porphyry (Eldursi K et al., 2018; Liu LM et al., 2016). Consequently, determining the spatial distribution of the monzonite porphyry is essential for lithological modeling and subsequent 3D geological reconstruction. This study employed the Petrel™ E&P software platform (developed by Schlumberger, Houston, Texas, USA) to model the 3D surface, structural, lithology, orebodies, and gold grades of the Tongshi Gold Deposit. The integrated workflow for 3D modeling and mineral prospectivity zone delineation is shown in Fig. 8.

(1) A 3D surface model was constructed based on the National Height Datum 1985 (NHD85). The model utilized the ordinary kriging interpolation method to process Gaofen-7 (GF-7) satellite-derived elevation data, integrating with geological maps, topographic maps, and borehole collar coordinates. Iterative calibration was performed in regions with conflicting field-mapped geological boundaries to ensure model fidelity.

(2) A 3D stratigraphic model was constructed using a convergent interpolation simulation based on borehole data. Fault distributions were mapped from exploration cross-sections, geological and topographic maps, and electromagnetic (EM) inversion results to construct the 3D fault model.

(3) The monzonite porphyry phases were correlated stratigraphically through well-log analysis. Isopach thickness contour maps of the monzonite porphyry were delineated based on the stratigraphic correlation results. Digitized maps were imported into the Petrel™ E&P software platform (Schlumberger Ltd.) to define the geological pinch-out boundaries, thereby constraining the spatial distribution of the monzonite porphyry. A structural grid framework (20 m × 20 m laterally, 1 m vertically) was established, followed by geostatistical modeling integrating variogram analysis (Genton MG, 1998; Li XH et al., 2018; Shafayi SH and Torab FM, 2022), deterministic interpolation, and sequential indicator simulation (Bohling G, 1994; Shafayi SH and Torab FM, 2022; Wang HC et al., 2022) to generate the lithology model.

(4) Orebody boundaries were delineated by vertically discretizing the study area into 1 m intervals using gold assay data. The boundary delineation adhered to the borehole spacing requirements specified in GB/T 13908-2020 (General Specifications for Solid Mineral Exploration) (Li HD et al., 2024; Nielsen RL, 2003). Subsequently, a 3D orebody model was generated using a deterministic interpolation method (Justine BG et al., 2024; Rossi M et al., 2017).

(5) The 3D gold grade model was constructed using variogram analysis of gold assay data that was constrained by lithology and orebody models (Genton MG, 1998; Li XH et al., 2018; Shafayi SH and Torab FM, 2022) and ordinary kriging interpolation (Huang et al., 2020; Xiao et al., 2015).

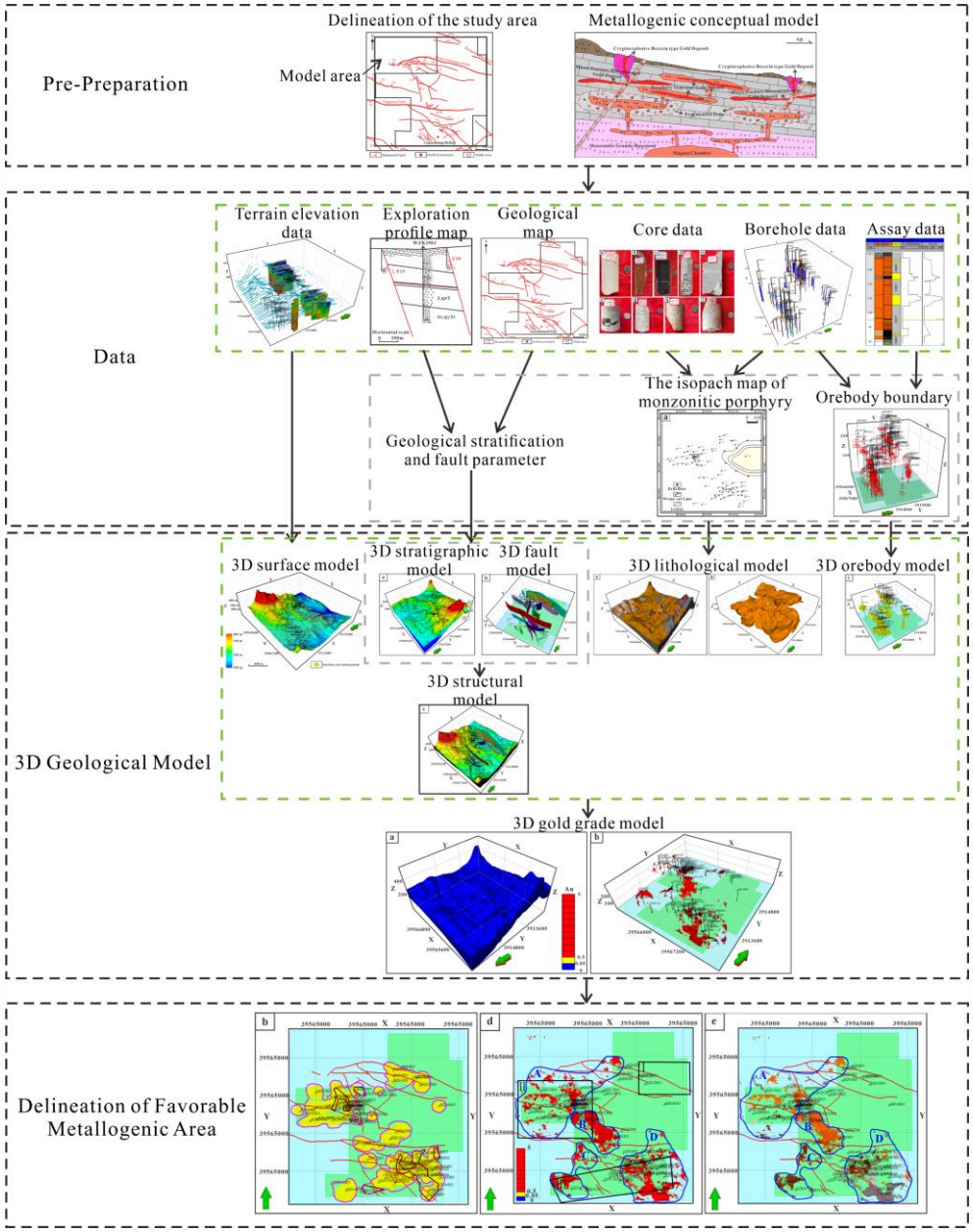


Fig. 8. The workflow for 3D geological modeling and prediction of favorable metallogenic areas.

5. Results

5.1. 3D Surface Model

The surface model is a digital representation of the topographic features of the mining area and can also show the shape and plane positions of the faults and other geological features. During 3D surface modeling, elevation data from the mining area were used to construct the initial 3D surface model. However, the localized areas lacked elevation data, which required that they be

refined. To refine the map, a 1:5000-scale topographic map and borehole collar coordinates were integrated as spatial constraints. An ordinary Kriging interpolation method was used to reconstruct the optimized 3D surface model (Fig. 9). The 3D surface model showed that the terrain in the study area is highest in the southwest and lowest in the northeast, and the highest point is located at Shiliugu Hill in the southern region.

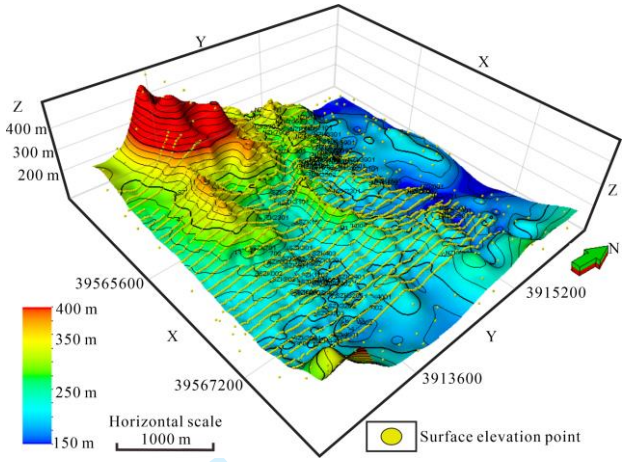


Fig. 9. Surface model of the study area derived using Petrel™ E&P software.

5.2. 3D Structural Model

Borehole-derived stratigraphic data were imported into the Petrel™ E&P software platform (Schlumberger Ltd.). A sequential indicator simulation (SIS) method was implemented, constrained by spatial trends from the surface model, to construct the stratigraphic model. Due to the higher concentrations of stratigraphic control points in the NW and SE sectors, stratigraphic inversion artifacts (older strata overlying younger units) occurred in data-sparse regions. To mitigate this, iterative adjustments were made using comprehensive horizon datasets and thickness variation trends to generate an optimized 3D stratigraphic framework model (Fig. 10a). Fault planes were extracted from topographic and geological maps, and stratigraphic profiles. Deterministic modeling based on surface-model-defined fault intersections weas used to construct the 3D fault model (Fig. 10b). When combined with the stratigraphic model, horizontal and vertical grids were generated to develop the 3D structural model (Fig. 10c).

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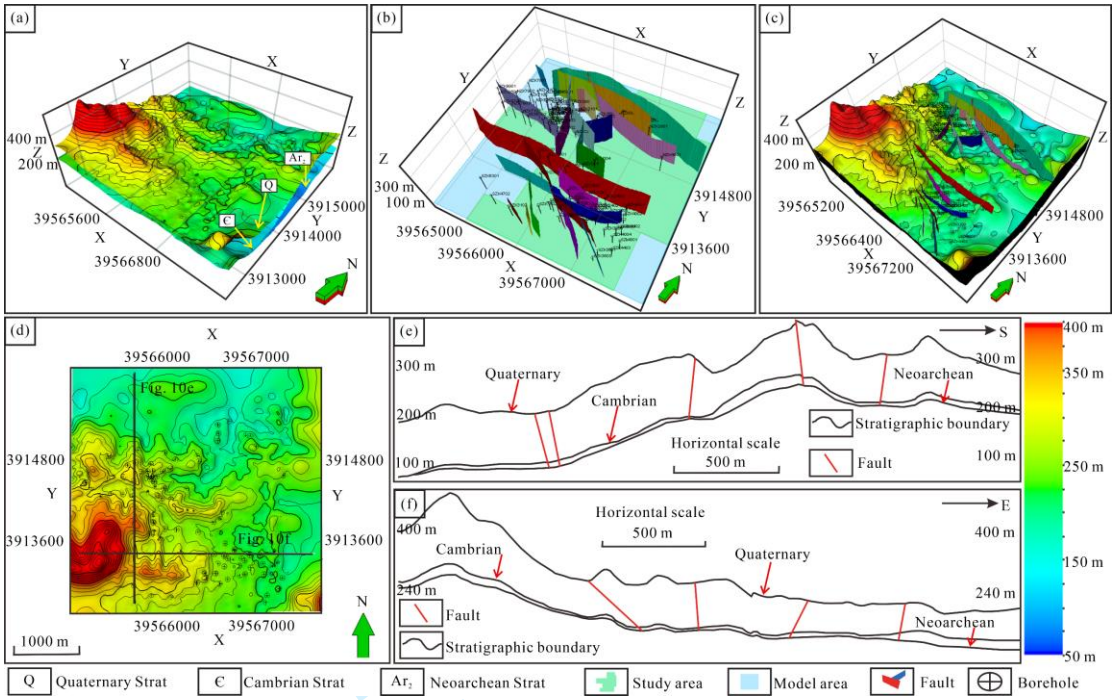


Fig. 10. 3D structural model of the study area derived using Petrel™ E&P and CorelDRAW software. (a) 3D stratigraphic model; (b) 3D fault model; (c) Structural framework model of the study area; (d) Locations of cross-sections; (e) North-south cross-section; (f) East-west cross-section.

The formations in the study area are mainly Quaternary, Cambrian, and Neoarchean strata. The Cambrian strata are exposed in the northwestern sector and the Neoarchean strata are exposed in the northeastern sector. The model contained 25 faults, mainly EW-trending faults. The dip angles were mainly between 60° and 80°, and the fault length was between 100 and 250 m. The Yingziwa fault is the main EW-trending fault, which is a tensile fault. The exposed length is about 5.20 km, the dip angle is 65°–86°, and the fault exhibits a sinuous geometry along the strike. These factors significantly influence the structural characteristics of the study area, and the development of the orebody and gold grade models.

5.3. 3D Lithology Model

The lithology model describes the 3D distribution of each type of rock, including the critical relationship between the orebody and host rock. The conventional methodology involves importing lithofacies data from core analysis or well log interpretation into the Petrel™ E&P software platform (Schlumberger Ltd.) and applying sequential indicator simulation (SIS) to generate a 3D lithology model (Jafarzadeh et al., 2019; Ren et al., 2019). However, the models generated via this method lack accuracy and fail to capture the spatial distribution patterns of the monzonite porphyry. To address these limitations, this study implemented a hybrid framework combining the deterministic interpolation method with sequential indicator simulation to reconstruct a geologically constrained 3D lithology model.

By adopting stratigraphic correlation methodologies analogous to hydrocarbon reservoir

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characterization (Fang XX et al., 2024; Fawad N et al., 2023; Zecchin M and Catuneanu O, 2015), intrusive phases of the monzonite porphyry were classified based on their distance to basement contact as the correlation criterion. Seventeen isopach thickness contour maps delineating the monzonite porphyry units were generated to define the spatial distribution of the intrusions (Fig. 11). A 3D lithology model (Fig. 12a) was constructed through deterministic interpolation and sequential indicator simulation following the anisotropy analysis of variogram functions. This modeling methodology employed deterministic constraints for lithological boundary delineation, with sequential indicator simulation applied to characterize the internal heterogeneity within the lithological boundaries. The lithology model generated via the deterministic interpolation method and SIS demonstrated significant advancements over conventional approaches that rely on stochastically generated entity models. This methodology effectively avoided the geologically inconsistent distribution patterns of monzonite porphyry, which are typically obtained when applying conventional modeling algorithms.

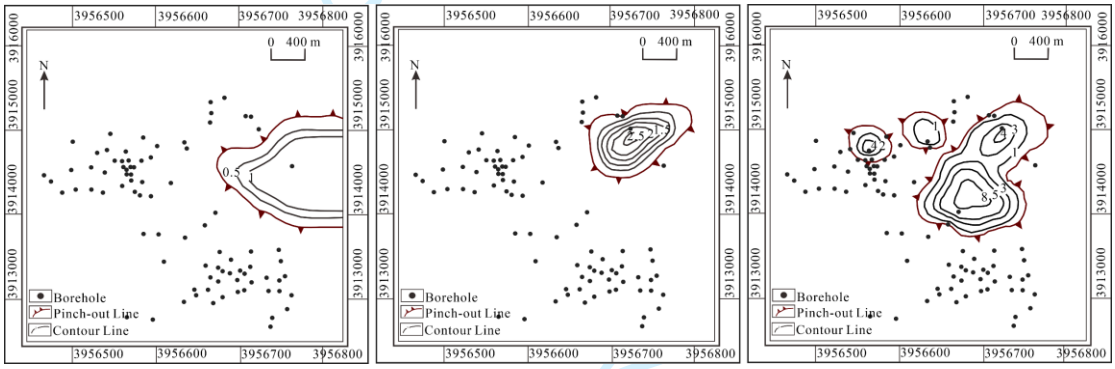


Fig. 11. Examples of isopach maps of monzonite porphyry in the study area derived using DF-GVision and CorelDRAW software.

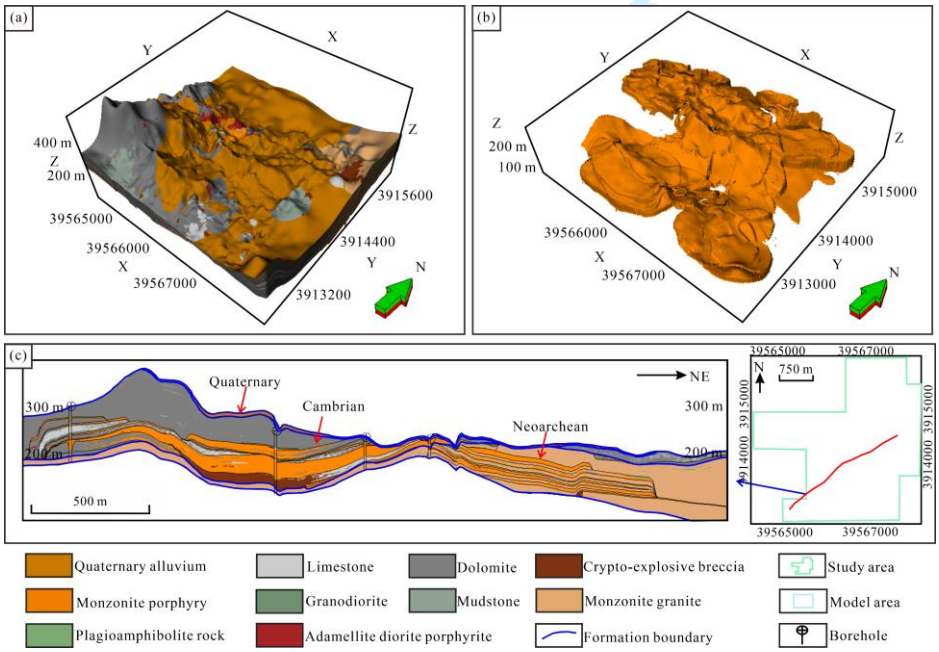


Fig. 12. The 3D lithology model of the study area derived using Petrel™ E&P software. (a) 3D

lithology model; (b) 3D monzonite porphyry model; (c) Lithologic borehole profile.

The lithology model indicates that monzonite porphyry units have stratiform-type geometries in cross-sectional profiles, while being distributed in clustered aggregates in plan-view (Figs. 11 and 12). The model also shows that monzonite porphyry predominantly occurs within Cambrian and Neoarchean strata in the central, western, northeastern, and southeastern parts of the study area, with thicknesses ranging from 1.40 m to 41.39 m.

5.4. 3D Orebody Model

The general industrial index of oxidized ore contained in Appendix D of "Specifications for Rock Gold Mineral Exploration" (Wang HY et al., 2020) indicates that the minimum gold concentration for a geobody to be considered an orebody is 0.5 g/t. To precisely describe the boundaries of the orebodies in the model, 192 equidistant elevation planes (1 m spacing) were vertically partitioned the study area into 191 intervals. Guided by the geological extrapolation criteria from GB/T 13908-2020 (General Specifications for Solid Mineral Exploration), mineralization properties were upscaled from borehole-scale observations to regional-scale models (Li HD et al., 2024; Nielsen RL, 2003). By integrating fault distributions and orebody occurrences identified in wells, the spatial distribution of the gold deposit was mapped in sections (Fig. 13a), and the orebody boundaries were delineated (Fig. 13b). The 3D orebody model was constructed based on the orebody boundary delineation results (Fig. 13c).

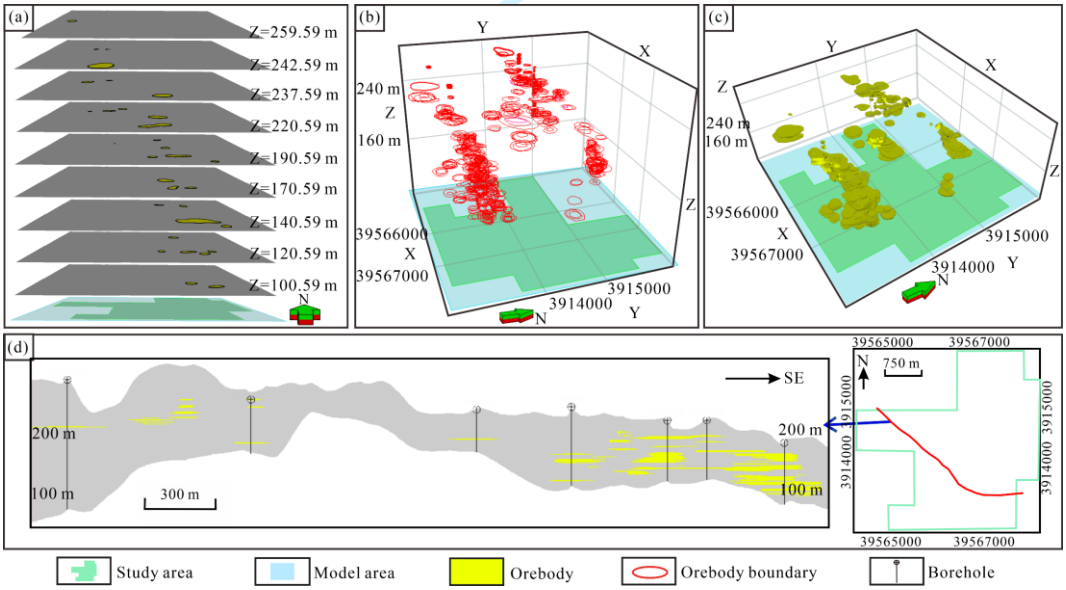


Fig. 13. The results of the 3D orebody model in the study area derived using Petrel™ E&P software. (a) Orebody plan–view distribution map; (b) Orebody boundaries; (c) 3D orebody model; (d) Cross-section of the study area.

The orebody model showed that the orebodies are mainly distributed in the central, western, and southeastern parts of the study area, with minor clusters occurring in the northeast. Vertically, the orebodies were stratified. The orebodies in the west are relatively shallow, with a depth range of 10.31 m to 136.58 m below the surface. The orebodies in the northeast are buried deeper, with a

depth range of 53.00 m to 141.80 m below the surface. The orebodies in the southeast are the most deeply buried, with a depth range of 9.34 m–157.34 m below the surface (Figs. 13 and 15).

5.5. 3D Gold Grade Model

Analysis of sample magnetic susceptibility data from the Tongshi mining district indicated that sedimentary rocks exhibit nonmagnetic to weakly magnetic properties (magnetic susceptibility range: 0–33 × 10^{-6} SI), while the magmatic rocks, such as the monzonite porphyry, have significantly higher magnetic susceptibilities (34–530 × 10^{-6} SI), due to their magnetite-rich mineral assemblages (Li PF, 2018). High-resolution magnetic survey data were acquired in August 2021 by Beijing Tanchuang Resources Technology Co. Ltd. delineated the high-magnetic susceptibility anomalies within the mining district. These anomalies were attributed to magnetite-bearing lithologies, predominantly monzonite porphyry-hosted hydrothermal breccias. The occurrences of gold mineralization in this zone were inferred to be genetically associated with the monzonite porphyry intrusions and were likely controlled by magmatic and hydrothermal fluid flow pathways through brittle-ductile shear zones (Liu DF, 2010; Zhang YM et al., 2024). The geological map of the exploration area, metallogenic model, geophysical surveys, and borehole datasets indicate that the gold orebodies occur in the Mesozoic monzonite porphyry and in the zones in which carbonate strata of the Cambrian Zhangxia Formation are in contact with the monzonite porphyry.

Analysis of the metallogenic model and magnetic anomaly characteristics demonstrated that the primary lithologies hosting gold mineralization included monzonite porphyry, carbonate rocks, and breccia. Therefore, constrained by the 3D lithology model, variogram analysis of the gold grades was performed on individual lithological units, followed by the construction of a 3D gold grade model (Fig. 14) using ordinary kriging interpolation.

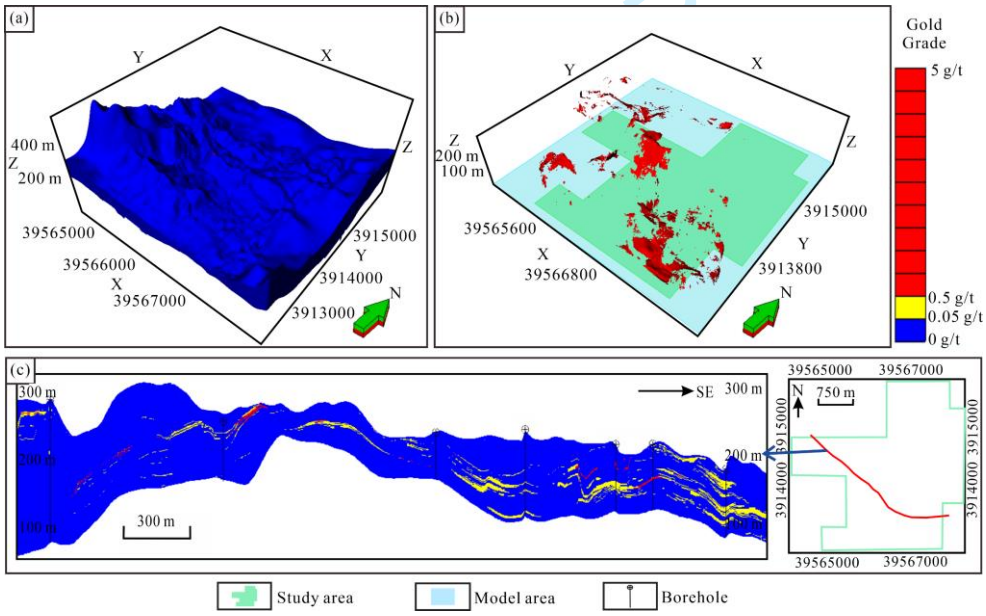


Fig. 14. The results of the 3D gold grade model in the study area derived using Petrel™ E&P

software. (a) 3D gold grade model; (b) 3D gold grade model (gold grades > 0.5 g/t); (c) Cross-section of the study area.

Previous studies have constructed 3D orebody models through explicit modeling approaches using borehole data and exploration cross-sections (Wang C et al., 2019; Zhang YM et al., 2024). Subsequently, the measured borehole grade data were interpolated within these orebody frameworks via kriging methods to generate a 3D grade model (Mao XC et al., 2014; Wang LL et al., 2024; Yu PP et al., 2019). However, these 3D grade models lacked constraints from the geological metallogenic model. This study integrated explicit and implicit modeling methodologies to develop a 3D lithology model within a structural grid framework. Subsequently, a 3D gold grade model was constructed under the constraints of the 3D lithology model. The 3D gold grade model integrated dual lithological and structural controls, reflecting the metallogenic model of the Tongshi Gold Deposit.

The resulting 3D model and the well profiles (Fig. 14c) indicate that the maximum gold grade is 5.34 g/t, which occurred in the western part of the study area, and gold ore was predominantly distributed in the western, central, and southern parts of the study area. In the central study area, gold occurred in a relatively shallow position with a depth range of 1.67 m–15.66 m below the surface. In the west and south, gold occurred in deeper intervals with a depth range of 10.31 m–157.34 m below the surface.

5.6. Model Validation

To validate the accuracy of the 3D gold grade model, three methodologies were implemented: correlation analysis, probabilistic distribution consistency evaluation, and borehole sampling validation. The measured gold grade data from boreholes and the corresponding simulated model data were analyzed using these approaches to assess the model's spatial distribution consistency and predictive capability.

The measured gold grade data from the boreholes and the corresponding simulated values in the model were statistically fitted (Fig. 15), revealing a strong correlation (Pearson's $r = 0.911$) between the modeled and actual gold grades. Statistical validation confirmed the model's reliability, consistent with previous studies demonstrating that correlation analysis effectively quantifies prediction accuracy through training dataset alignment (Chen J et al., 2024; Koike K et al., 2022; Zheng CJ et al., 2023).

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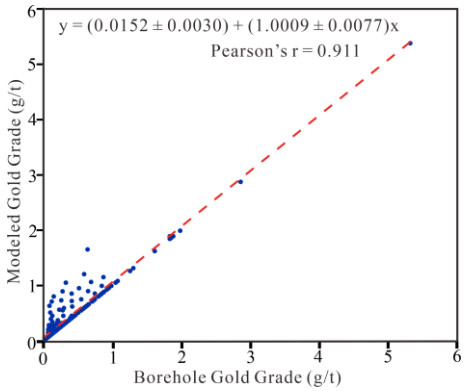


Fig. 15. Linear correlation between borehole and modeled gold grades in 3D geological modeling derived using Petrel™ E&P and OriginPro software.

The probability distribution consistency assessment was conducted to analyze the similarity of the distribution patterns between the original borehole gold grade data and the modeled values. Higher consistency in key parameters, including primary distribution intervals and peak values, indicates a stronger model accuracy (Hou WS et al., 2017; Wang MC et al., 2018; Zhang ZQ et al., 2023). Frequency distribution analysis (Fig. 16) demonstrated a close correspondence between the modeled and borehole gold grades in dominant ranges and peak values, validating the model's capacity to replicate the statistical properties of the original dataset.

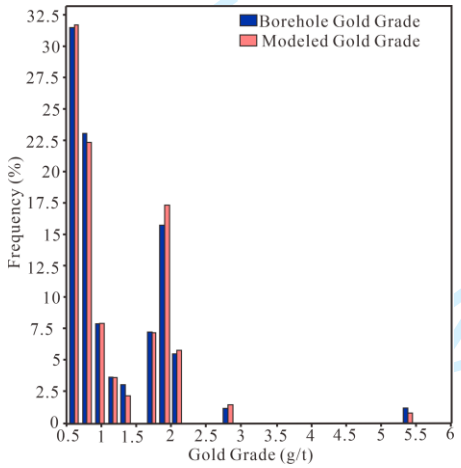


Fig. 16. Probability distribution of gold grade data derived using Petrel™ E&P software.

To validate the 3D geological model through borehole exclusion, four boreholes (S01, S04, S07, and S09) were excluded from the modeling process (Deng CS et al., 2023). A comparative analysis between the original gold grade data from these excluded boreholes and the 3D gold grade model outputs (Fig. 17) revealed significant consistency in the spatial distribution patterns, confirming the accuracy of the reconstructed geological framework.

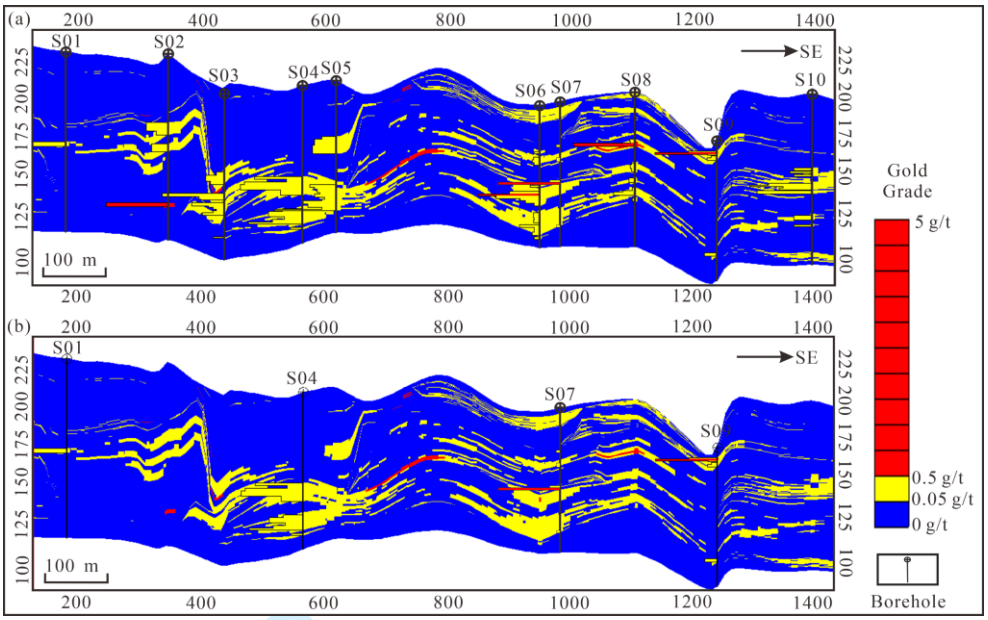


Fig. 17. Comparative analysis of 3D gold grade cross-sections derived using Petrel™ E&P software. (a) 3D gold grade model cross-sections; (b) Validation cross-sections: original borehole data excluded from modeling vs. 3D geological model.

5.7. Delineation of Favorable Metallogenic Area

Based on the 3D gold grade model, four favorable mineralization zones were delineated using the defined industrial gold grade boundary of 0.5 g/t as the constraint (Wang HY et al., 2020). In area A, gold mineralization mainly occurred in the deep monzonite porphyry and shallow limestone and dolomite in contact with the monzonite porphyry. In area B, gold was hosted in the shallow monzonite porphyry. The gold in the area C occurred in the breccia in contact with the monzonite porphyry in shallow intervals. In area D, gold occurred in deep Cambrian limestone and dolomite. The fault systems in the area exhibit layered distributions, which is consistent with the microfracture-disseminated gold orebodies predicted here (Fig. 18).

The magnetic anomaly properties of the three exploration targets acquired in previous studies are consistent with the favorable metallogenic areas delineated by 3D modeling technology (Li XH et al., 2018; Mao GY et al., 2015; Zhang YM et al., 2024). Targets II and III represented prospective layered microfracture-disseminated zones (Fig. 18b). The distribution of the monzonite porphyry controls these mineralization patterns, which show a strong spatial correspondence with the predicted Zones A and D (Fig. 18b).

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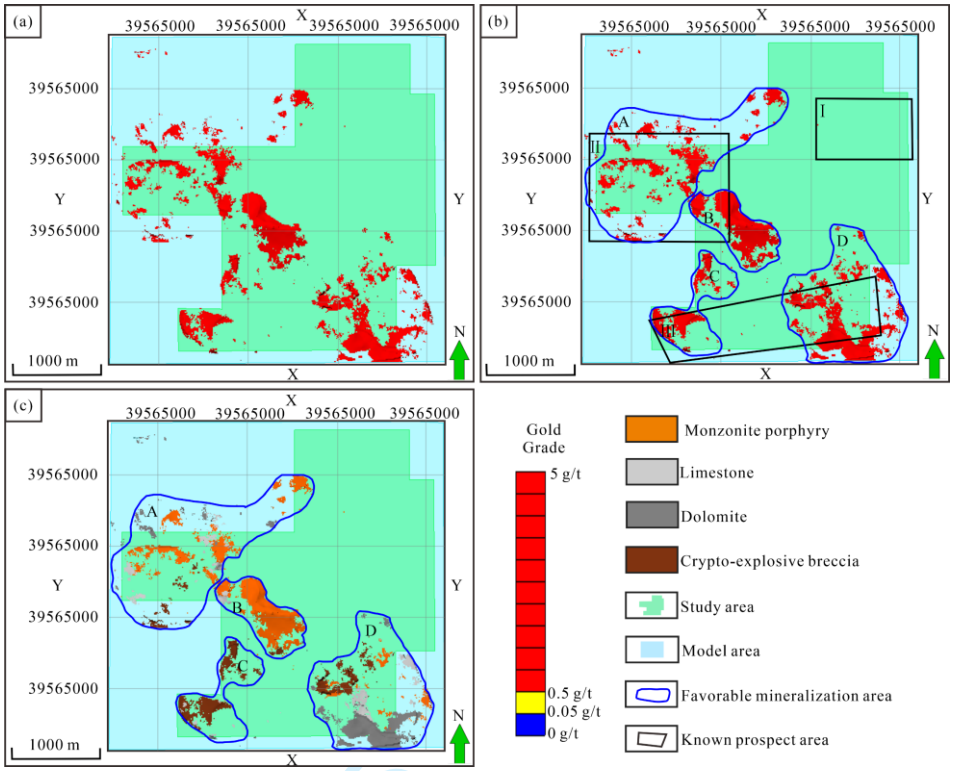


Fig. 18. Prediction of favorable metallogenic areas in the study area derived using Petrel™ E&P and CorelDRAW software. (a) The gold grade distribution plan-view (gold grades > 0.5 g/t); (b) Prospective metallogenic zones; (c) The lithology of the prospective metallogenic areas.

6. Discussion

6.1. Relationship between Gold Deposit and Faults

The Yingziwa Fault has a structural partitioning effect on the distribution of gold deposits, so the study area can be subdivided into South and North Zones along its trace (Fig. 19). Interrogating the results of the model shows that the northern gold orebodies were formed in close association with monzonite porphyry distributions, and that fault controls were secondary. The gold orebodies were predominantly hosted in the monzonite porphyry and the limestone dolomite zones in contact with the monzonite porphyry (Huang JN et al., 2010; Figs. 15 and 16).

In the southern zone, the formation of gold orebodies was mainly influenced by faults and microfractures. The southeastern sector has a high fault density, with NW-trending- and sub-E–W-trending faults radiating outward, resulting in abundant microfractures. These faults and microfractures provided migration channels and structural traps for ore-bearing hydrothermal fluids, directing fluid flow toward the reactive limestone and dolomite layers. The reaction of hydrothermal fluids with dolomite limestone led to the formation of microfracture-disseminated gold orebodies. Due to the tectonic, stratigraphic, and magmatic conditions in the southeast, the gold orebodies were more extensively developed and deeply buried than in the southwestern sector (Figs. 13c and 20).

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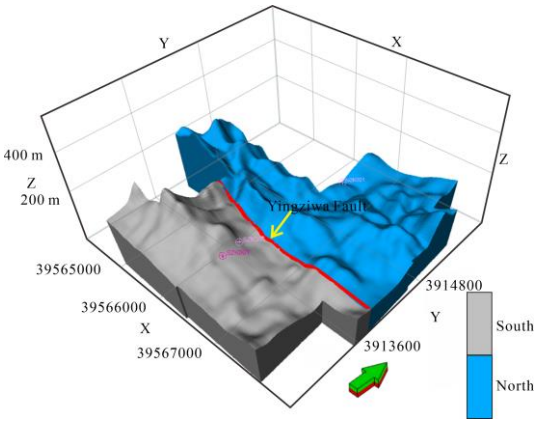


Fig. 19. The Northern Zone and the Southern Zone in the model derived using Petrel™ E&P software.

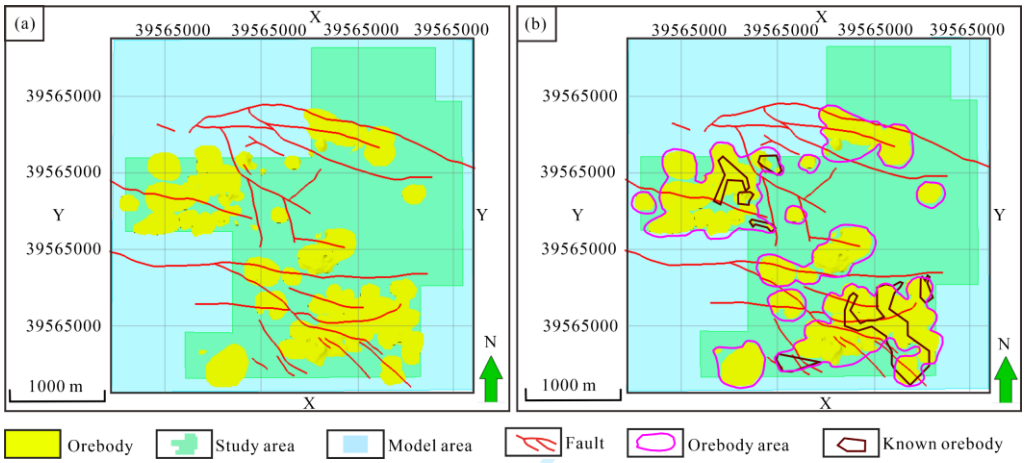


Fig. 20. Planar distribution and predictive map of orebodies in the study area derived using Petrel™ E&P software. (a) Plan-view of orebody distribution; (b) Plan-view of orebody prediction.

6.2. Relationship between Gold Deposit and Lithology

The formation of gold deposits is closely related to the rock type. The 3D model showed that the gold-bearing rocks were mainly limestone, dolomite, crypto-explosive breccia, and monzonite porphyry (Fig. 18c). The gold-bearing monzonite porphyry is dominated by medium- to coarse- and coarse-spotted porphyrites, which are developed in the eastern, northern, and central regions (Fig. 21). The gold-bearing limestone and dolomite rocks occur predominantly within their contact zones with the monzonite porphyry, and the fault and microfracture densities are enhanced relative to the adjacent domains. Faults and microfractures provide migration channels for ore-bearing hydrothermal fluids, facilitating fluid-rock reactions with limestone and dolomite. The gold-bearing crypto-explosive breccias are mainly developed near the Yingziwa Fault, Mawoyindonggou Fault, Fault F12 (Fig. 3), and in the contact zone with the monzonite porphyry. The gold-bearing rocks in the breccia were mainly monzonite porphyry breccia.

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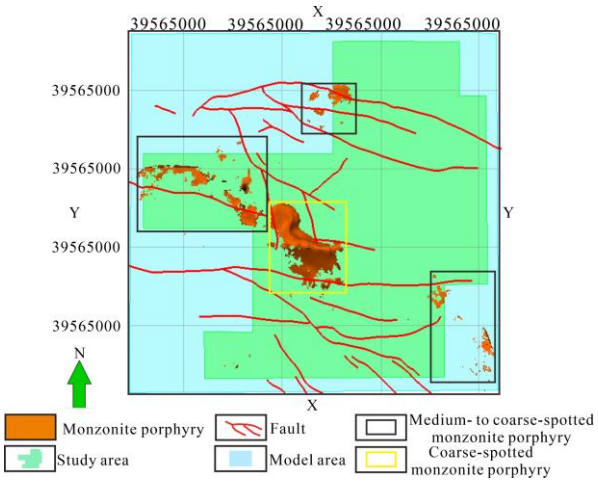


Fig. 21. Distribution of gold-bearing monzonite porphyry.

Statistical analysis of the gold assay data showed decreasing mean and peak gold grades across lithologies: monzonite porphyry > crypto-explosive breccia > dolomite > limestone. The maximum gold grade of monzonite porphyry is 5.34 g/t, with an average of 0.22 g/t (Fig.22a). According to core observations, the monzonite porphyry was classified into fine-, medium- to fine-, medium-, medium- to coarse-, and coarse-spotted monzonite porphyry. The medium- to coarse-spotted monzonite porphyry exhibited the highest mean (0.50 g/t) and peak (5.34 g/t) grades in the monzonite porphyry (Fig. 22b).

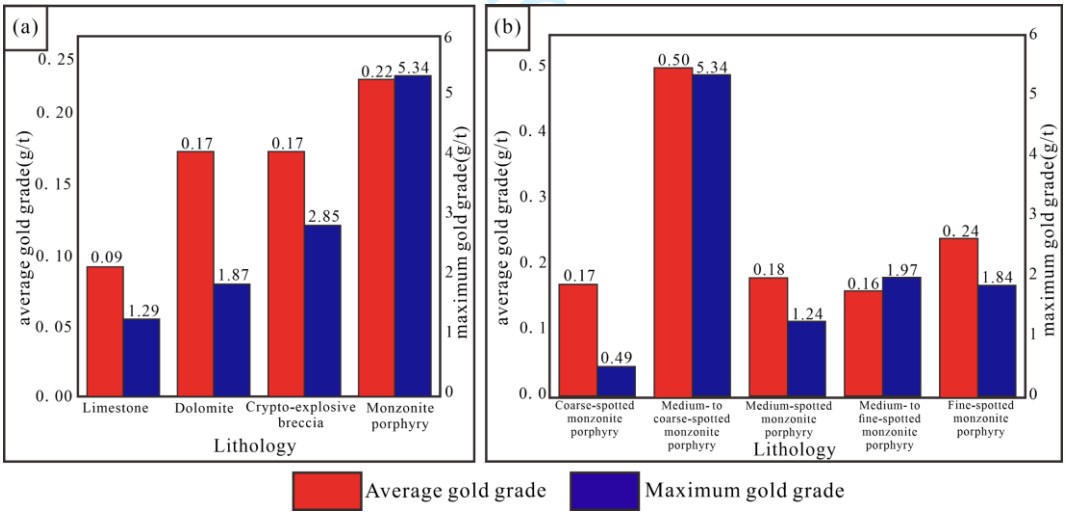


Fig. 22. Relationship of the gold grade to lithology in the study area derived using OriginPro software. (a) Diagram of the gold grades of different lithologies; (b) Diagram of the gold grade of different monzonite porphyry samples.

6.3. Indicators of Orebody Occurrences

By integrating the data from the different sources, including the stratigraphic conditions, structural conditions, magmatic rock conditions, metallogenic model, and 3D geological model, prospecting criteria for gold deposits can be summarized as follows:

(1) The gold content of the medium- to coarse-spotted monzonite porphyry is high,

suggesting favorable exploration targets. Mineralized bodies are predominantly localized within the monzonite porphyry contact zones, aligning with the metallogenic model. The reactive carbonate lithologies (limestone and dolomite) in these zones enhance fluid-rock interactions and serve as critical lithological indicators of exploration prospects within the study area.

(2) Faults and microfractures provided migration channels and structural traps for ore-bearing hydrothermal fluids. The radial distribution of the NW-trending and sub-E–W-trending faults is conducive to the formation of gold orebodies. The radial fault systems and fault intersection areas serve as critical structural indicators of possible exploration prospects within the study area.

(3) Crypto-explosive breccias are closely related to the mineralization and occurrence of gold, in which gold enrichment correlates with breccia fragment size and matrix content. Large breccia fragments with a low matrix content generally have low porosity, which inhibits gold enrichment. Conversely, small breccia fragments with high matrix content have enhanced permeability, which favors gold enrichment. These characteristics can prioritize exploration prospects in zones of dense crypto-explosive breccias, particularly veined types along fault zones or intersections.

7. Conclusions

The previous studies in the Tongshi Gold deposit area primarily utilized 2D methodologies, which could not adequately delineate the complex geological processes and orebody distributions. In this study, we integrated the metallogenic model of the deposit with a 3D geological model to elucidate the spatial relationships between the orebody, geological structures, and stratigraphy, refine and validate conclusions derived from prior 2D studies, and derive conclusions only attainable through 3D analysis, thereby providing a robust framework for prospecting in analogous metallogenic settings.

(1) Integrated analysis of Mesozoic magmatic activity, fault-fracture systems, and carbonate stratigraphy, supported by geochemical data, shows epithermal gold deposits (low-temperature, shallow hydrothermal systems). Three metallogenic models are developed: crypto-explosive breccia-type, microfracture-disseminated-type, and porphyry-vein-type gold deposits. These models indicate that gold mineralization predominantly occurs as stratiform bodies within monzonite porphyry and its contact zones with limestone and dolomite strata. The genetic linkage of gold mineralization with monzonite porphyry directly guides 3D geological modeling of gold deposits.

(2) By integrating multiple datasets, including exploration cross-sections, drill core data, and gold grade assays, a 3D geological model was constructed. The 3D model and previous results demonstrate that the orebodies occur mainly in the monzonite porphyry and the contact zones of limestone and dolomite layers with the monzonite porphyry. Consequently, the area of monzonite porphyry development in the Mesozoic subvolcanic complex and its contact boundary with the Cambrian Zhangxia Formation limestone strata are the most prospective intervals for the

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monzonite porphyry altered-type gold deposits and the layered microfracture-disseminated-type gold deposits.

(3) The factors controlling ore occurrence in the Tongshi Gold Deposit in Liujiazhuang Beiling area comprise three key components: 1) lithological controls—reactive limestone and dolomite in contact with medium- to coarse-spotted monzonite porphyry; 2) structural controls—radial fault systems and fault intersection zones; and 3) other mineralization features—crypto-explosive breccias along fault zones.

(4) The southeastern sector of the study area has favorable stratigraphic, structural, and magmatic controls, resulting in enriched gold mineralization. The burial depths of the orebodies in this zone range from 9.34 m–157.34 m below the surface. In the central part of the study area, gold occurs in a relatively shallow position, with a depth range of 1.67 m–15.66 m below the surface. In the west and south, the gold occurs in the deepest intervals, with a depth range of 10.31 m–136.58 m below the surface.

(5) This research develops an innovative methodology integrating the Petrel™ E&P software platform 3D visualization and multiple sources of data to enhance model construction and identification of prospective areas for gold deposits controlled by hydrothermal systems.

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Declarations

Conflict of Interest The authors declare that they have no known competing financial interests or personal relationships.

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